

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5

8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

EER MODEL: GENERALIZATION AND SPECIALIZATION WITH A UNIVERSITY CASE STUDY

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INTRODUCTION TO EER MODEL

EER is an extension of the traditional ER Model

Helps represent complex real-world database structures

Includes inheritance, specialization, and generalization

Improves database design accuracy and flexibility

Widely used in universities, hospitals, and banking systems

Supports superclass and subclass concepts

Bridges the gap between conceptual and logical design



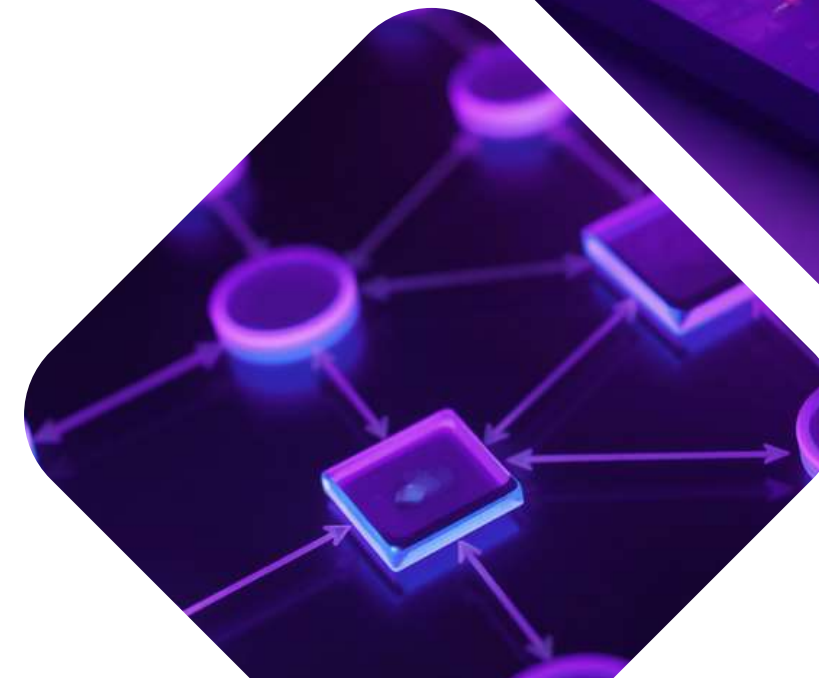
What is the EER Model?

Definition & Core Concepts

The Enhanced Entity Relationship (EER) Model extends the traditional ER model with advanced features to represent complex database structures more accurately.

- Represents entities, attributes, and relationships more clearly
- Supports superclass and subclass concepts (inheritance)
- Efficiently models real-world objects and scenarios
- Simplifies database management and maintenance

EER bridges the gap between real-world complexity and database design — making it ideal for university, hospital, and banking systems.



FEATURES OF THE EER MODEL

Supports Generalization and Specialization

Uses inheritance between entities

Handles complex relationships easily

Reduces data redundancy

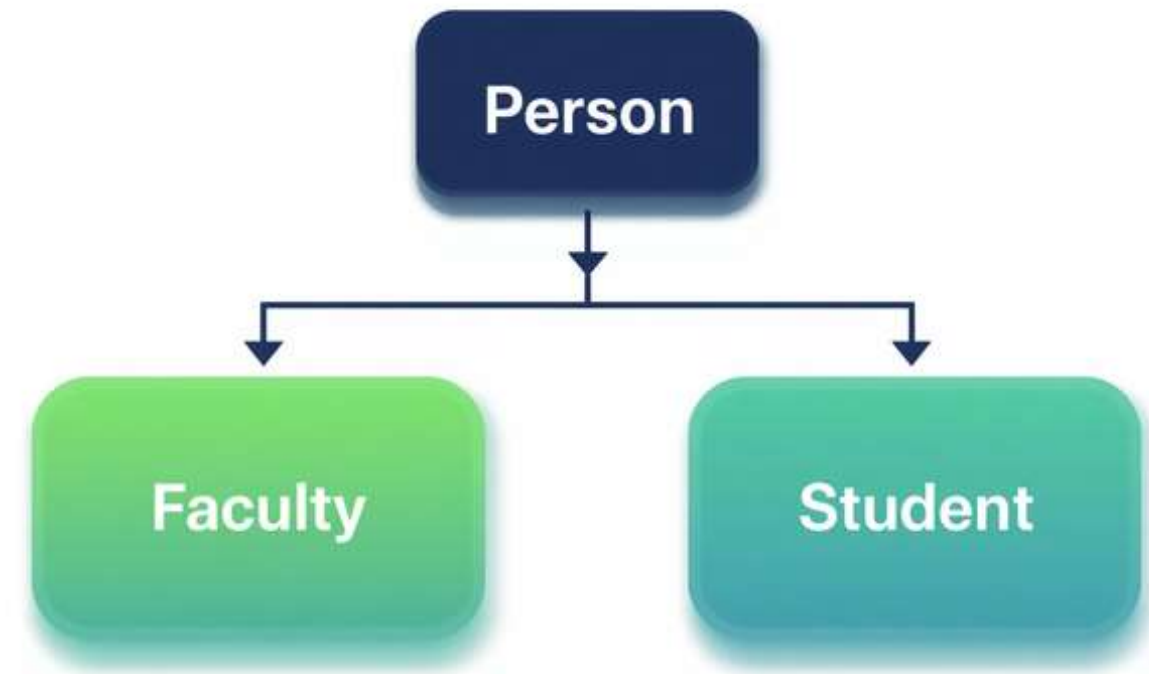
Improves database organization



GENERALIZATION

What is Generalization?

Generalization is a bottom-up approach in the EER Model that combines multiple similar entities into a single higher-level superclass. Common attributes shared by all entities are stored only once in the superclass, reducing data duplication. This process creates a more organized and consistent database structure, improving overall database design accuracy and flexibility.



Example: Faculty + Student → Person

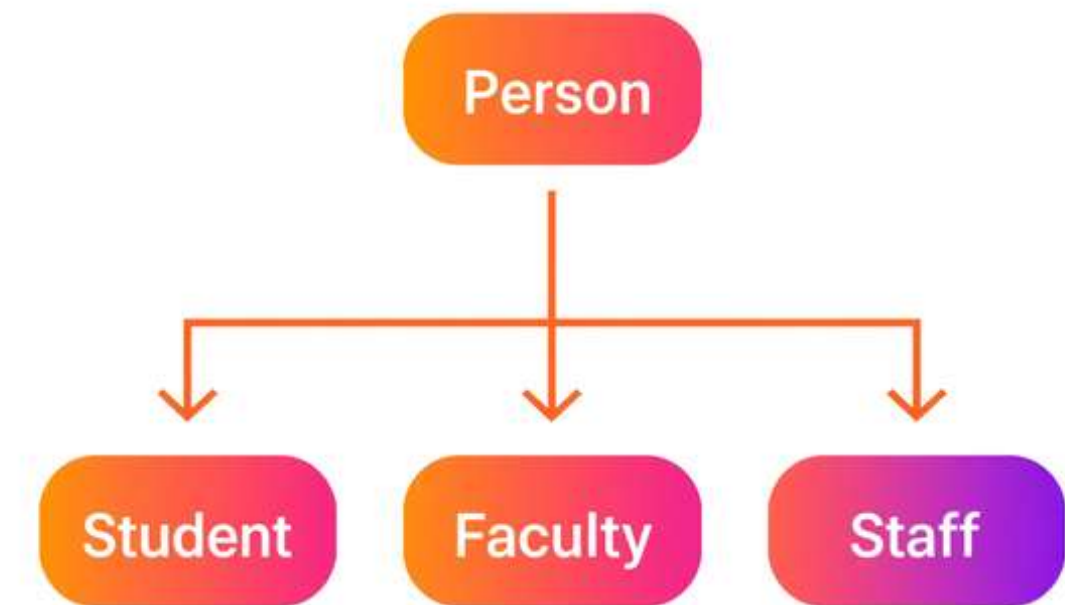
In a university database, Faculty and Student share common attributes such as Person ID, Name, Address, Phone, and Email. By generalizing these into a Person superclass, redundancy is eliminated. Both Faculty and Student inherit from Person, while retaining their own unique attributes. This approach improves database consistency and simplifies maintenance.

SPECIALIZATION

What is Specialization?

Specialization is a top-down approach in the EER Model where a higher-level entity (superclass) is divided into smaller, more specific subclasses. Each subclass inherits common attributes from the superclass and also contains its own unique attributes, providing more detailed and accurate information.

Example: Person → Student, Faculty



Key Characteristics

- Divides one entity into smaller subclasses
- Each subclass has unique attributes
- Provides more detailed information
- Supports inheritance from superclass
- Improves data accuracy and flexibility
- Subclasses can have their own relationships



GENERALIZATION VS SPECIALIZATION

01

Generalization (Bottom to Top)

- Combines similar entities into one superclass
- Creates a higher-level entity
- Removes redundancy
- Stores common attributes only once
- Example: Faculty + Student → Person

02

Specialization (Top to Bottom)

- Divides one entity into smaller subclasses
- Creates detailed subclasses
- Adds unique attributes per subclass
- Supports inheritance from superclass
- Example: Person → Student, Faculty

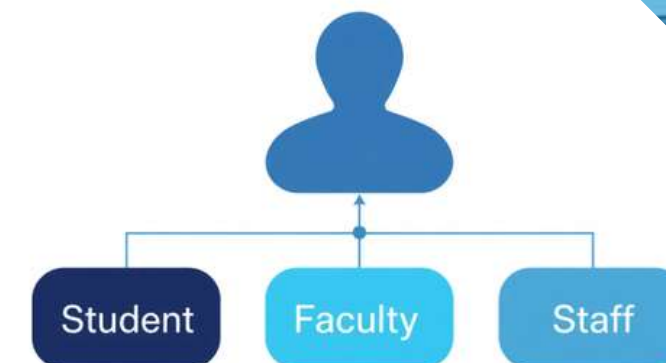
University Case Study

Introducing the University Context

A university environment manages three key groups: Students, Faculty, and Staff. Each group shares common personal information, making them ideal candidates for a unified superclass in the EER Model.

In this case study, all three groups are modeled under a single Person superclass, which stores shared attributes such as Person ID, Name, Address, Phone Number, and Email — eliminating redundancy across the database.

Student, Faculty, and Staff are defined as subclasses that inherit all common attributes from the Person superclass, while each subclass also maintains its own unique attributes specific to its role in the university system.



SUPERCLASS: PERSON



✓ 01

Person ID and Name

Every person in the university system is uniquely identified by a Person ID. The Name attribute stores the full name and is inherited by all subclasses: Student, Faculty, and Staff.

✓ 02

Address and Phone Number

The Address attribute captures the residential or official location of each person. Phone Number provides a direct contact reference, shared across all person types in the university database.

✓ 03

Email

The Email attribute serves as the primary digital communication channel. All five attributes — Person ID, Name, Address, Phone Number, and Email — are common to every person entity and are stored only once in the superclass to avoid redundancy.



SUBCLASSES

01

Student

Roll Number: Unique identifier for each student

Course: Program enrolled (e.g., B.Sc CS)

Semester: Current semester of study

Faculty

Employee ID: Unique faculty identifier

Department: Assigned department (e.g., CS)

Salary: Monthly compensation details

02

Staff

Staff ID: Unique staff member identifier

Job Role: Position held (e.g., Admin, Lab Tech)

Office: Office location or room number

Note: Each subclass inherits common attributes (Person ID, Name, Address, Phone, Email) from the Person superclass and adds its own unique attributes.

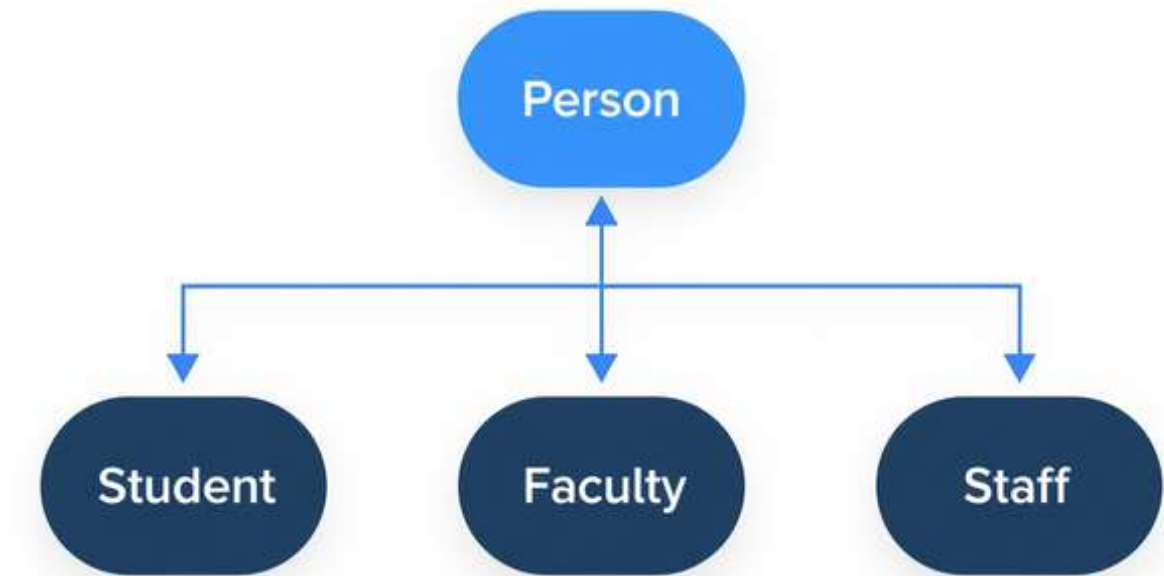
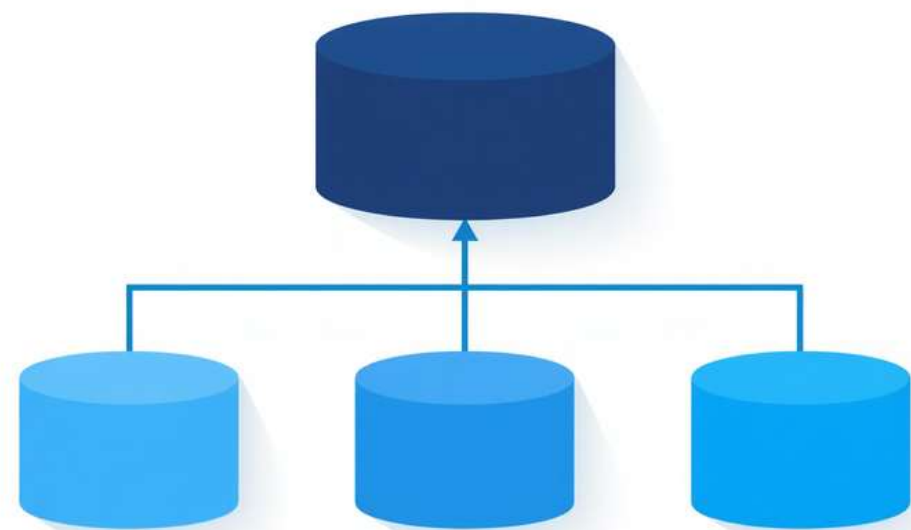
UNIVERSITY EER DIAGRAM

Superclass: Person

Person is the superclass containing common attributes shared by all university members:

- Person ID
- Name
- Address
- Phone Number
- Email

Inheritance arrows connect Person to subclasses: Student, Faculty, and Staff — avoiding duplicate data and improving database organization.



Subclasses & Inheritance

Student: Roll Number, Course, Semester

Faculty: Employee ID, Department, Salary

Staff: Staff ID, Job Role, Office

All subclasses inherit Person attributes. This hierarchy eliminates redundancy, supports scalability, and accurately models the university database structure using EER specialization.

ADVANTAGES OF EER MODEL

- Reduces duplicate data storage
- Improves database flexibility
- Represents real-world objects clearly
- Simplifies database maintenance
- Supports future database expansion
- Enhances inheritance and reusability
- Improves overall database organization



Database Advantage

It is a good place to store data. It is a good place to store data. It is a good place to store data. It is a good place to store data. It is a good place to store data.



Logical Advantage

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Physical Advantage

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Logical Advantage

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Physical Advantage

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APPLICATIONS

University Management Systems

Hospital Information Systems

Banking Databases

Library Management Systems

Employee Management Systems





CONCLUSION

- EER is an advanced extension of the traditional ER model with powerful features.
- Generalization combines similar entities (e.g., Student + Faculty → Person) into a superclass.
- Specialization divides a superclass into detailed subclasses with unique attributes.
- The University case study clearly demonstrates inheritance among Person, Student, Faculty, and Staff.
- EER significantly improves database design, reduces redundancy, and enhances management.



REFERENCES

- [1] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed. Boston, MA, USA: Pearson, 2017.
- [2] A. Silberschatz, H. F. Korth, and S. Sudarshan, Database System Concepts, 7th ed. New York, NY, USA: McGraw-Hill, 2020.
- [3] C. J. Date, An Introduction to Database Systems, 8th ed. Pearson Education, 2004.
- [4] T. Connolly and C. Begg, Database Systems: A Practical Approach to Design, Implementation, and Management, 6th ed. Pearson, 2015.
- [5] P. Rob and C. Coronel, Database Systems: Design, Implementation, and Management, 13th ed. Cengage Learning, 2019.





THANK YOU!

